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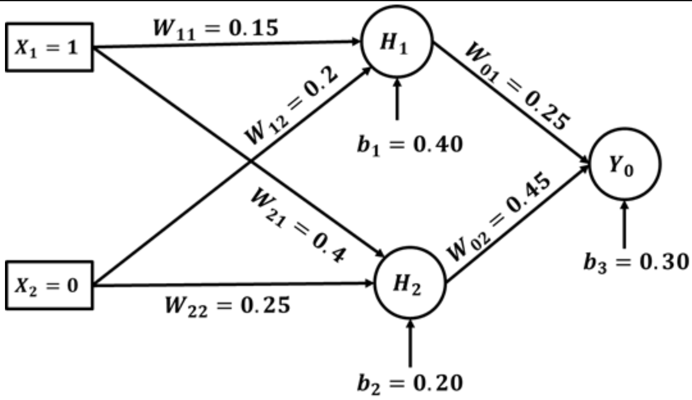
Question bank
CSA3007 Deep learning

UNIT	Question Description
1	Analyse the difference between underfitting and overfitting of Machine learning model.
	Describe the significant role of Eigen values, Eigen Vectors and Eigen Decomposition in Deep learning.
	What is the role of stride and padding in the convolution operation, and how do they affect the output feature map size?
	Explain the concepts of Entropy, Cross-Entropy, and KL Divergence with mathematical formulations. Derive the relationship between them and evaluate their role in training deep learning model.
	A medical test is used to detect a certain disease. The following probabilities are known: • $P(Disease) = 0.01$ • $P(No\ Disease) = 0.99$ • $P(Positive\|Disease) = 0.96$ • $P(Positive\|No\ Disease) = 0.04$ (a) Using Bayes' Theorem, compute: $P(Disease \mid Positive)$ (b) Explain the concept of Bayesian Classification. Clearly define: • Prior probability • Likelihood • Posterior probability (c) Explain how Bayesian Classification uses prior and likelihood probabilities in decision making.
	Explain overflow and underflow problems in numerical computation. Discuss their impact on deep learning models and describe methods to overcome them.
	A dataset contains 100 samples used to train a binary classifier for detecting a particular class. Out of these, 60 samples belong to Class A and 40 belong to Class B. A new instance has the following likelihoods: • P(X|A)=0.7 • P(X|B)=0.2 1. Using Bayes' Rule, calculate the posterior probabilities P(A|X) and P(B|X) 2. Based on the result, determine which class the instance belongs to using Bayesian classification. If the model has very high capacity and perfectly fits the training data but performs poorly on new data, identify whether the model suffers from overfitting or underfitting and justify your answer.
	Analyze the impact of overflow and underflow in numerical computation, explain how gradient-based optimization methods are used in building machine learning algorithms, compare supervised and unsupervised learning, and explain the role of Bayesian classification in machine learning models with examples.

	Analyze the role of linear algebra concepts such as vectors, matrices, and tensors in the implementation of deep learning models, and Explain the role of probability theory in machine learning and illustrate how it is used for modeling uncertainty and making predictions.
	The first bag contains 3 white balls, 2 red balls and 4 black balls. Second bag contains 2 white, 3 red and 5 black balls and third bag contains 3 white, 4 red and 2 black balls. One bag is chosen at random and from it 3 balls are drawn. Out of three balls two balls are white and one is red. What are the probabilities that they were taken from first bag, second bag and third bag.
	Explain gradient-based and constrained optimization techniques in deep learning, and analyze their influence on the learning process and model performance. Examine the role of eigen decomposition and matrix norms in feature extraction and dimensionality reduction, and discuss how these support supervised and unsupervised learning paradigms with their limitations.
	Explain tensors and their role in deep learning architectures, and discuss how numerical issues such as overflow and underflow affect model computations and performance.
	Explain the concept of eigenvalues and eigenvectors and discuss their importance in Deep Learning. Also, find the eigenvalues and corresponding eigenvectors of the following matrix: $A = [2\ 1\ 0\ 0\ 3\ 1\ 0\ 0\ 4]$
	$P(A) = 0.6\ , P(B) = 0.5, P(B\|A) = 0.7$ 1. Use Bayes' Theorem to compute $P(A\|B)$. 2. Interpret the result in the context of classification. 3. Evaluate the importance of Bayesian Classification in machine learning. 4. Discuss issues of overflow and underflow in numerical computation.
	Compare and contrast over-fitting with under-fitting from the following aspects. a. Definition b. Identifying each of them c. Common causes d. Training error, testing error, model complexity e. Fixing each of the errors
	How does the gating mechanism in an LSTM help address the vanishing gradient problem observed in standard RNNs?
	What are the key differences between an RNN and a feedforward neural network in terms of architecture and functionality?
2	Compare and contrast LSTMs and GRUs in terms of structure, performance, and computational efficiency.
	For a Multi-Layer Perceptron (MLP) network as shown in the diagram, given the initial weights, biases, and input values, the network uses a sigmoid activation function for both hidden and output neurons. For the target output t = 1 and learning rate η=0.1, perform one iteration of the delta learning rule for the output layer to update the output weights W_{01}, W_{02} and bias b_3.



	Explain the architecture and working of Convolutional Neural Networks (CNNs), describing the convolution operation, filters, padding, stride, pooling layers, dropout and fully connected layers. Also, discuss the applications of CNNs.
	Describe the perceptron model and its learning process in ANN, highlighting its role as a linear classifier in high-dimensional feature spaces. Relate this to the importance of pooling and dropout in CNN, showing how they enable hierarchical feature abstraction and improve generalization by reducing overfitting.
	Explain the Convolutional Neural Network (CNN) with the help of the following terminologies. A. Why are CNNs preferred over traditional ANNs for image data? B. What are the main layers in CNN? C. What is a convolution operation? What is a feature map?
	Consider the cost function used in linear regression: $J(w) = (w - 3)^2$ Let the initial weight be $w_0 = 6$. Using Gradient Descent, perform three iterations of the update rule for each of the following learning rates: $\eta = 0.1, \eta = 0.5, \text{ and } \eta = 1.2$ For each learning rate, compute and find the updated weights w_1, w_2, and w_3. Based on the obtained values of w_1, w_2, and w_3, compare the convergence behavior for different learning rates and explain what happens when the learning rate is too small or too large.
	Explain the architecture of an LSTM network and its functionality in detail.
	Explain the concept of attention models and discuss how they improve the performance of encoder–decoder architectures.
	Analyze the working of Artificial Neural Networks (ANN) by explaining the perceptron model, learning rules, network layers, and the backpropagation algorithm, and examine how popular architectures, parallel and distributed processing (PDP), linear associative models, and stochastic networks enhance the learning capabilities of neural networks.
	Explain the architecture and functioning of Convolutional Neural Networks (CNN) by discussing convolution, filters, pooling, stride, dropout, layers, and their applications. Additionally, describe the structure of Recurrent Neural Networks (RNN), emphasizing unfolding and the Backpropagation Through Time (BPTT) algorithm used for training on sequential data.
	Evaluate the contribution of LSTM networks and bidirectional neural networks in improving the performance of deep learning models for complex sequence prediction tasks.

	Explain the architecture and learning mechanism of Artificial Neural Networks (ANN), including perceptron, learning laws, and backpropagation, and analyze their limitations in complex learning tasks. Further, examine how Parallel and Distributed Processing (PDP), linear associative models, and stochastic networks enhance learning capability and overcome these limitations.
	Explain the working principles of Convolutional Neural Networks (CNN), including convolution, filters, pooling, stride, and dropout, and analyze their impact on feature extraction. Relate these concepts to Recurrent Neural Networks (RNN), explaining unfolding, Backpropagation Through Time (BPTT), LSTM, and bidirectional networks for sequential data processing.
3	Compare VAEs and GANs in terms of their strengths and weaknesses. In what scenarios would you prefer one over the other?
	Explain Deep Boltzmann Machines (DBM). How do they differ from Restricted Boltzmann Machines (RBM)?
	Compare the working mechanisms of Variational Autoencoder, Deep Autoencoder, and Generative Adversarial Network with suitable diagram and examples.
	Analyze the role of Deep Neural Networks (DNNs) in handling sequence and streaming data. Examine how deep learning models can be used for real-time multimedia processing and prediction tasks.
	Describe the architecture and learning mechanism of Variational Autoencoders (VAE) and Deep Autoencoders (DAE), and compare their roles in representation learning and generative modeling. Relate these models to Generative Adversarial Networks (GAN), highlighting differences in training dynamics, output quality, associated challenges, and applications in multimedia data.
	Discuss the structure and functioning of Deep Boltzmann Machines (DBM) and Deep Belief Networks (DBN) in probabilistic learning. Show how these models capture complex data distributions and support unsupervised learning.
	Discuss the working principle of Generative Adversarial Networks (GAN) in terms of generator–discriminator interaction and adversarial training. Relate this to the role of deep learning models in multimedia and sequence data processing, highlighting their ability to learn complex representations.
	Explain the concept of a Variational Autoencoder (VAE) as a generative model. In your answer, discuss: 1. The difference between a traditional autoencoder and a variational autoencoder. 2. The VAE loss function, including the reconstruction loss and KL-divergence term. 3. Applications of VAEs in generative modelling and representation learning.
	Given a dataset of images with noise: 1. Explain how a Deep Autoencoder (DAE) can be used for noise reduction. 2. Outline the steps involved in training the model. 3. Mention suitable activation functions and loss functions.
	Explain Deep Boltzmann Machines (DBM). How do they differ from Restricted Boltzmann Machines (RBM)?
	What are the key differences between the encoder and decoder in an autoencoder in terms of function and architecture?
	How does the adversarial training process in a GAN differ from traditional supervised learning?
	Explain the architecture and working of a Generative Adversarial Network (GAN). In your answer, discuss the following:

	1. Describe the GAN architecture, clearly explaining the roles of the Generator and Discriminator and their interaction during training. 2. Explain the objective function and loss function of GAN, including the min-max optimization problem with mathematical formulation. Discuss different types of GANs and their applications.
	Explain the architecture and working of a Deep Autoencoder (DAE). In your answer, discuss the following: 1. The difference between a shallow autoencoder and a deep autoencoder. 2. The role of multiple hidden layers in feature extraction and representation learning. 3. The concept of encoding and decoding in deep autoencoders. 4. How deep autoencoders can be used for dimensionality reduction and data compression.
	Explain the concept of optimization in deep learning. Discuss the challenges in neural network optimization such as vanishing gradient, exploding gradient, and saddle points, and their impact on model training.
	Why do deep neural networks suffer from the vanishing gradient and exploding gradient problems, and how do they impact training?
	Explain the concept of unsupervised feature learning in Convolutional Neural Networks (CNNs) and describe how CNNs learn hierarchical feature representations from data.
	Discuss the challenges involved in optimizing neural networks, and describe basic optimization algorithms as well as adaptive learning rate methods used for training deep models.
	Discuss the motivation behind Convolutional Networks, their structured output mechanisms, unsupervised feature learning, and the neuroscientific principles that inspired their design.
	Discuss the different gradient descent optimization techniques such as batch, stochastic, and mini-batch gradient descent used for efficient training of deep learning models, with examples.
	Examine the challenges in optimizing deep neural networks, including vanishing gradients and non-convex loss functions, and evaluate the effectiveness of basic and adaptive optimization algorithms. Connect these methods with approximate second-order techniques and meta-optimization strategies for improving convergence and training efficiency.
4	Analyze the motivation behind Convolutional Networks and their role in structured output learning, along with the significance of unsupervised feature extraction, linking these concepts to neuroscientific principles that influenced their design and functioning.
	Outline basic and adaptive optimization algorithms used in deep learning. Link these with optimization strategies and meta-algorithms for enhancing model training performance.
	Design a strategy to optimize the training of a deep convolutional neural network for image classification. Your design should include: • Choice of optimizer • Learning rate strategy • Handling overfitting • Use of unsupervised features • Role of neuroscientific inspiration in CNN design
	Explain the optimization challenges, such as vanishing gradients, exploding gradients, and saddle points. Analyze their impact on deep learning models.
	Explain the working of basic optimization algorithms such as Gradient Descent, Stochastic Gradient Descent (SGD), with the help of an example.
	Why is adaptive learning rate optimization (e.g., Adam, Adagrad, and RMSprop) beneficial for training deep neural networks?

	How does the choice of activation function affect optimization, and why are ReLU and its variants preferred over sigmoid and tanh in deep networks?
	Describe various optimization strategies and meta-algorithms used in deep learning. Explain techniques such as learning rate scheduling, early stopping, regularization, and batch normalization.
	Describe the architecture and working of Recurrent Neural Networks (RNNs). Explain their advantages and limitations in handling sequential data.
	Difference between classification vs regression loss functions in deep learning.
	"Recurrent Neural Networks (RNNs) are widely used for processing sequential data such as text, speech, and time-series information." Explain the architecture and working of Recurrent Neural Networks and justify how Bidirectional RNN improves sequence learning.
	"Echo State Networks provide an efficient approach to training recurrent neural networks by keeping most of the network weights fixed." Discuss the working principle of Echo State Networks and justify their significance in sequence modeling tasks.
	Discuss the criteria for selecting machine learning or deep learning approaches for large-scale tasks, illustrated with case studies in classification and regression.
	Describe the structure and functioning of Echo State Networks and outline their advantages over traditional RNN models, linking their efficiency to improved training dynamics. Associate these concepts with practical methodologies in large-scale deep learning systems, including scalability and optimization, highlighting their applications in classification, regression, and deep network-based problem solving.
	Discuss computational graphs and their role in automatic differentiation for training deep neural networks. Describe the architecture of Recurrent Neural Networks (RNN), including bidirectional and deep recurrent networks for sequence modelling, and highlight their significance in handling temporal dependencies in large-scale applications.
5	Analyze the differences between: • RNN vs Deep RNN • RNN vs Echo State Network Discuss advantages, limitations, and use cases.
	Given a simple RNN with an input sequence $x_1 = 1, x_2 = 2$, weight $w = 0.5$, and the initial hidden state $h_0 = 0$: 1. Compute the hidden states h_1 and h_2 using: $$h_t = w \cdot x_t + w \cdot h_{t-1}$$ 2. Show all steps. 3. Explain how previous information influences the output.
	Evaluate the effectiveness of Recurrent Networks in large-scale deep learning applications, such as: • Speech recognition • Text generation Discuss strengths and limitations.

	Explain bidirectional RNN and its advantages over standard RNN, highlighting improved context capture from past and future sequences. Combine this with applications of deep learning in classification and regression tasks, showing how such architectures enhance prediction accuracy and performance.
	How do Bidirectional RNNs (BiRNNs) differ from traditional unidirectional RNNs in terms of capturing sequential information, and why is this important for certain tasks?
	How do Deep Recurrent Networks (DRNs) differ from traditional shallow RNNs, and what advantages do they provide in modeling complex sequential dependencies?
	How does linear regression differ from logistic regression in terms of the type of problem they solve and the underlying assumptions about the data?
	Explain the concept of Bidirectional Recurrent Neural Networks (BiRNNs) and analyze how they improve upon standard RNNs. Discuss with suitable examples.